DEMAND FORECASTING AND OPERATIONAL ANALYSIS FOR A HOME-BASED CLOUD KITCHEN

A Final report for the BDM capstone Project

Submitted by

Name: M.Hema Varshni

Roll number: 24f1001256

IITM Online BS Degree Program,

Indian Institute of Technology, Madras, Chennai

Tamil Nadu, India, 600036

Contents

1 Executive Summary 3

2 Detailed Explanation of Analysis Method 4

2.1 Data Collection and Preprocessing 4

2.2 Demand Forecasting Methods 4

2.3 Inventory Optimization Methods 5

2.4 Menu Profitability Analysis 7

2.5 Procurement Cost-Benefit Analysis 9

2.6 Python-Based Data Analysis (Google Colab) 10

3 Results and Findings 10

3.1 Daily Order Volume Trend 10

3.2 Day-of-Week Demand Pattern 11

3.3 Order Type Distribution 12

3.4 Meal Period Analysis 12

3.8 Statistical Distribution and Outlier Analysis 15

4 Interpretation of Results and Recommendations 16

4.1 PS1 – Emergency Procurement Reduction 16

4.2 PS2 – Peak Hour Capacity Management 17

4.3 PS3 – Menu Profitability Optimization 17

5 Dataset Links and References 19

Declaration Statement

I am working on a Project titled **"Demand Forecasting and Operational Analysis on a Home-based Cloud Kitchen"**. I extend my appreciation to **Padma's Home Food**, for providing the necessary resources that enabled me to conduct my project.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered from primary sources and carefully analyzed to assure its reliability.

Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the principles of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I understand that all recommendations made in this project report are within the context of the academic project taken up towards course fulfillment in the BS Degree Program offered by IIT Madras. The institution does not endorse any of the claims or comments.

Signature of Candidate: (**Digital Signature**)

Name: M. Hema Varshini

Date: 12-10-2025

1 Executive Summary

Padma's Home Food is a cloud kitchen run by Mrs. Padmavathy Ravichandran through the Cookr app in Medavakkam, Chennai. Operating since February 2023 on a B2C model, it serves customers within a 2–3 km radius with over 30 authentic home-style Indian dishes across breakfast, lunch, and dinner, offered through both on-demand orders and monthly subscription plans. Despite strong ratings and consistent demand, the business faces three critical challenges: ingredient stockouts from unpredictable demand forcing emergency purchases at 20–30% markup (PS1); peak-hour surges — especially Thursday lunchtimes — causing cook exhaustion and unplanned closures (PS2); and a lack of dish-level profitability visibility that leaves high-margin opportunities unrealised while loss-making dishes persist in the menu (PS3).

This study analyses 567 orders over 49 operational days (April 1 – May 26, 2025), supplemented by structured interviews on costs and procurement. Four datasets were built — Order Data, Menu Details, Ingredient Costs, and Operational Events — and analysed using demand forecasting (7-Day Moving Average and Exponential Smoothing), inventory tools (ABC Classification, EOQ, ROP), menu engineering (Contribution Margin matrix), and procurement cost comparisons, all validated in Excel and Python.

Key findings show high demand volatility (CV = 44.44% for orders; CV = 53.03% for revenue), with Thursday as the peak day (100 orders, 17.6% of weekly volume) and lunch accounting for 74.1% of all orders. Regular orders generate 68.8% of revenue from just 47.4% of order volume (AOV: Rs. 302 vs. Rs. 123 for subscriptions). Menu engineering of 20 dishes identifies four Stars (including Chicken Thokku at 45.8% CM and White Rice at 49.4%), one underpriced Plowhorse (Chicken Biryani at 29.2% CM), and one loss-making Dog (Ven Pongal at –1.8% CM). Tomato and Chicken carry Quick Commerce markups of 140% and 80% respectively, underscoring the cost of unplanned procurement.

Three interconnected interventions are recommended, targeting an estimated annual improvement of Rs. 4,13,000–4,25,000. A Sunday forecast review with structured Monday–Tuesday bulk procurement and ROP-triggered restocking of 7 Class A ingredients can eliminate 70–80% of emergency purchases, saving Rs. 18,000–30,000 annually (PS1). Daily order caps (15 on Thursday–Friday; 18 on Monday–Wednesday) and cross-training a kitchen helper can eliminate unplanned closures and recover Rs. 40,000 in lost revenue (PS2). Repricing Chicken Biryani to Rs. 210, removing Ven Pongal from subscriptions, and

prioritising Star dishes can lift the average contribution margin from 32% to 42%, generating Rs. 51,840 in additional annual contribution (PS3).

2 Detailed Explanation of Analysis Method

2.1 Data Collection and Preprocessing

Order data was sourced directly from Mrs. Padmavathy's Cookr app history, manually transcribed into Google Sheets, and imported into Excel. The dataset covers 567 orders across 49 operational days (April 1–May 26, 2025), capturing weekday/weekend patterns, the Tamil New Year period, summer holidays, and a 7-day May closure due to illness. Transcription accuracy was validated via an 8.8% random audit (50 records) cross-referenced against app screenshots.

Four datasets were created: **Order Data** (567 rows, 10 columns), **Menu Details** (264 rows covering prices, prep times, and ingredients), **Ingredient Costs** (87 ingredients with local vs. Quick Commerce pricing), and **Operational Events** (15 entries flagging festivals, holidays, and closures).

Preprocessing involved consolidating 47 raw dish name variants into 32 canonical names, standardising date formats, validating data types, imputing subscription order values, and creating a derived TYPE_FIXED column to fill blank order-type fields. All steps are documented in the Excel Analysis Workbook (see Section 5).

2.2 Demand Forecasting Methods

Two complementary forecasting methods were applied to the 49-day daily order count time series. The 7-Day Moving Average provides a smoothed baseline that captures the weekly demand cycle inherent in the data, while Exponential Smoothing provides a more adaptive forecast that responds to recent demand shifts.

7-Day Moving Average (MA):

```
MA(t) = [D(t-1) + D(t-2) + ... + D(t-7)] / 7
```

where D(t) is actual daily order count on day t. A window of n = 7 was selected because the business operates on a weekly cycle with consistent day-of-week patterns (Thursday peak, Monday trough). The MA was computed from Day 8 onwards in the Demand Forecasting sheet of the Excel workbook.

Exponential Smoothing (ES):

$$F(t+1) = \alpha \times D(t) + (1 - \alpha) \times F(t)$$

where F(t+1) is the forecast for the next day, D(t) is the actual demand on day t, F(t) is the previous forecast, and α is the smoothing parameter. After testing α values of 0.1, 0.2, 0.3, and 0.4, α = 0.3 was selected as it minimises MAPE while retaining responsiveness to genuine demand shifts. The smoothing weight of 0.3 gives 30% weight to the most recent observation and 70% to the accumulated forecast history, appropriate for a dataset with moderate volatility.

Model Evaluation — Mean Absolute Percentage Error (MAPE):

$$MAPE = (1/n) \times \Sigma \, |D(t) - F(t)| \, / \, D(t) \times 100\%$$

The Exponential Smoothing model achieved a MAPE of 25.51% when evaluated on operational days (excluding 6 kitchen closure days where orders fell to 1–2 due to illness, not demand fluctuation). Including closure days inflates MAPE to 120.7%, which misrepresents forecast accuracy since closures are operational events, not forecastable demand patterns. For comparison, the 7-Day Moving Average achieved a MAPE of 22.4%. Both models are appropriate for the 3–7 day planning horizon required for procurement decisions, and both are Excel-implementable without requiring statistical software.

2.3 Inventory Optimization Methods

ABC Classification:

ABC classification was applied to 16 key ingredients to prioritise inventory management effort. Ingredients were ranked by annual consumption value (annual usage × unit cost), and cumulative percentage thresholds were applied: Class A (top 70% of annual value) requiring tight control and frequent review; Class B (next 20%) requiring moderate control; and Class C (bottom 10%) requiring minimal oversight.

Table 2.1 presents the full ABC classification. Seven ingredients fall into Class A: White Rice (Rs. 33,000/year), Toor Dal (Rs. 32,400/year), Chicken (Rs. 28,800/year), Ghee (Rs. 15,600/year), Cooking Oil (Rs. 14,400/year), Basmati Rice (Rs. 11,040/year), and Coconut (Rs. 9,600/year). These 7 ingredients represent 69.3% of total annual ingredient expenditure and are the primary focus of the EOQ and ROP calculations.

Ingredient	Local Cost (₹)	Monthly Usage	Units	Annual Value (₹)	Cum. %	Class
White rice	55	50	kg	33,000	15.8%	A
Toor dal	180	15	kg	32,400	31.3%	A
Chicken	200	12	kg	28,800	45.1%	A
Ghee	650	2	L	15,600	52.5%	A
Cooking oil	150	8	L	14,400	59.4%	A
Basmati rice	115	8	kg	11,040	64.7%	A
Coconut	50	16	pcs	9,600	69.3%	A
Curd	80	10	L	9,600	73.9%	B
Tamarind	400	2	kg	9,600	78.5%	B
Cashews	750	1	kg	9,000	82.8%	B
Eggs	6	120	pcs	8,640	86.9%	B
Onion	35	20	kg	8,400	90.9%	C
Tomato	25	20	kg	6,000	93.8%	C
Potato	30	15	kg	5,400	96.4%	C
Drumstick	15	30	pcs	5,400	98.9%	C
Sambar powder	45	4	packs	2,160	100%	C

Table 2.1: ABC Classification of Key Ingredients by Annual Consumption Value

Economic Order Quantity (EOQ):

$$EOQ = \sqrt{(2 \times D \times S / H)}$$

where D = annual demand (units), S = ordering cost per order (Rs.), and H = annual holding cost per unit (Rs./unit/year). EOQ determines the optimal order quantity that minimises the

total of ordering costs and holding costs. For Toor Dal: D = 182.5 kg/year, S = Rs. 50 per order trip, H = Rs. 18/kg/year (10% of cost), giving EOQ = √(2 × 182.5 × 50 / 18) = 31.84 kg per order, compared to the current ad hoc purchase of 5–10 kg, indicating significant potential for cost reduction through bulk ordering.

Reorder Point (ROP) and Safety Stock:

$$Safety\ Stock = Z \times \sigma_demand \times \sqrt{(Lead\ Time)}$$

$$ROP = (Avg\ Daily\ Demand \times Lead\ Time) + Safety\ Stock$$

where Z = 1.645 (for 95% service level), σ_demand = standard deviation of daily demand, and Lead Time is the procurement lead time in days. The ROP signals when stock has fallen to the level at which a new order must be placed to avoid stockout before the next delivery arrives. Table 2.2 presents the ROP and Safety Stock values for the 7 Class A ingredients.

Ingredient	Avg Daily (kg/pcs)	Std Dev	Lead Time (days)	Annual Demand	EOQ	Safety Stock	ROP
Toor dal	0.50 kg	0.2	2	182.5	31.84	0.47	1.47
White rice	1.67 kg	0.5	2	609.6	105.27	1.16	4.50
Chicken	0.40 kg	0.3	1	146	27.02	0.49	0.89
Ghee	0.07 L	0.03	3	25.6	6.27	0.09	0.30
Cooking oil	0.27 L	0.1	3	98.6	25.63	0.28	1.09
Basmati rice	0.27 kg	0.1	2	96	—	0.23	0.77
Coconut	0.53 pcs	0.2	2	192	—	0.47	1.53

Table 2.2: ROP and Safety Stock for Class A Ingredients (95% Service Level, Z = 1.645)

2.4 Menu Profitability Analysis

Contribution Margin Analysis:

$$Contribution\ Margin\ (CM) = Selling\ Price - Ingredient\ Cost - Platform\ Fee$$

$$CM\ Ratio = CM / Selling\ Price \times 100\%$$

Platform fee is fixed at 30% of selling price for all orders on the Cookr platform. Ingredient costs were estimated using actual local vendor prices from the Ingredient Costs dataset, with standard serving sizes verified with Mrs. Padmavathy. Gas/fuel cost of Rs. 5 per dish was included in the variable cost. Table 2.3 presents the CM analysis for the top 10 regular dishes and Table 2.4 for the top 10 subscription dishes.

Dish	Sell (₹)	Ing. Cost	Platform	Total Var.	CM (₹)	CM%	Category
Keerai paruppu kadayal	120	33.77	36.00	69.77	50.23	41.9%	Star
Chicken Biryani	190	77.56	57.00	134.56	55.44	29.2%	Plowhorse
Vegetable poriyal	65	40.40	19.50	59.90	5.10	7.8%	Plowhorse
Chicken thokku	160	38.72	48.00	86.72	73.28	45.8%	Star
White rice	80	16.50	24.00	40.50	39.50	49.4%	Star
Chicken 65	150	49.67	45.00	94.67	55.33	36.9%	Star
Rasam	65	20.95	19.50	40.45	24.55	37.8%	Puzzle
Keerai kootu	80	38.85	24.00	62.85	17.15	21.4%	Dog
Vazhakkai varuval	80	29.77	24.00	53.77	26.23	32.8%	Puzzle
Vendakkai poriyal	70	15.43	21.00	36.43	33.57	48.0%	Puzzle

Table 2.3: Contribution Margin Analysis — Top 10 Regular Dishes

Dish	Sell (₹)	Ing. Cost	Platform	Total Var.	CM (₹)	CM%	Category
Ragi idiyapam+white kurma	99	38.14	29.70	67.84	31.16	31.5%	Puzzle
Rice+Fish gravy+Rasam	179	74.63	53.70	128.33	50.67	28.3%	Plowhorse
Chicken fried rice+sauce	149	59.43	44.70	104.13	44.87	30.1%	Puzzle

Dish	Sell (₹)	Ing. Cost	Platform	Total Var.	CM (₹)	CM%	Category
Idly(4nos)+vada curry	99	32.66	29.70	62.36	36.64	37.0%	Puzzle
Rice+Dhal+Chicken65+Egg+Rasam	129	57.20	38.70	95.90	33.10	25.7%	Plowhorse
Ven pongal+Sambar+Coconut chutney	79	56.71	23.70	80.41	-1.41	-1.8%	Dog (LOSS)
Rice+Rasam+Chicken uppu kari	129	48.50	38.70	87.20	41.80	32.4%	Star
Lemon rice+Egg pepper fry	99	35.61	29.70	65.31	33.69	34.0%	Puzzle
Chapathi(4pcs)+Chicken curry	129	45.12	38.70	83.82	45.18	35.0%	Star
Pesarattu dosa+Chutney+Omelette	99	34.90	29.70	64.60	34.40	34.7%	Puzzle

Table 2.4: Contribution Margin Analysis — Top 10 Subscription Dishes

Menu Engineering Matrix:

Each dish was classified into one of four quadrants based on its relative popularity (order frequency) and CM ratio. Stars (high popularity, CM ≥ 30%) are strategic assets. Plowhorses (high popularity, CM < 30%) generate volume but constrain profitability. Puzzles (low popularity, CM ≥ 30%) have margin potential but require better promotion. Dogs (low popularity, CM < 30%) should be reviewed for removal or repricing.

2.5 Procurement Cost-Benefit Analysis

```
Markup % = (Quick Commerce Price - Local Vendor Price) / Local Vendor Price × 100%
```

This analysis compares the cost differential between emergency Quick Commerce procurement and planned local vendor purchasing across all 87 ingredients. Critical high-markup ingredients include: Tomato (140% markup: Rs. 25 local vs Rs. 60 on Quick Commerce), Chicken (80% markup: Rs. 200 vs Rs. 360), Drumstick (80%: Rs. 15 vs Rs. 27), Potato (66.7%: Rs. 30 vs Rs. 50), and Eggs (133%: Rs. 6 vs Rs. 14). Given that Mrs. Padmavathy currently sources tomatoes and chicken through Quick Commerce during demand surges, the annual saving from switching to planned procurement is estimated at Rs. 18,000–30,000.

2.6 Python-Based Data Analysis (Google Colab)

To validate the Excel findings and produce publication-quality charts, the order dataset was analysed in Python (Google Colab) using pandas for aggregation and matplotlib for visualisation, with a custom Exponential Smoothing loop implemented to independently verify the MAPE value. The notebook — available via the dataset link in Section 5 — produces five key outputs: a daily order volume trend chart with demand phases annotated, a day-of-week order volume bar chart, a meal category pie chart, an Exponential Smoothing forecast vs. actual comparison chart, and a subscription vs. regular revenue bar chart. All outputs correspond directly to the figures in Section 3.

3 Results and Findings

3.1 Daily Order Volume Trend

Daily order volume over the 49-day period reveals a clearly phased demand pattern that directly validates the unpredictability described in PS1. Phase 1 (April 1–15) recorded the highest average at 15.2 orders per day, driven by a new subscription cycle and Tamil New Year activity. Phase 2 (April 16 – May 10) stabilised at 12.3 orders per day as renewals moderated and regular orders settled into a weekly rhythm. Phase 3 (May 11–26) saw a sharp decline, with demand falling to near-zero during a 7-day kitchen closure due to illness (May 17–23), followed by a slow recovery.

The coefficient of variation for daily order volume was 44.44% against a mean of 11.81 orders per day — nearly half the mean — confirming that intuition-based procurement is structurally insufficient. The closure alone caused an estimated revenue loss of Rs. 9,718 (7 days × mean daily revenue of Rs. 1,388), with an additional 5–7 day recovery lag as subscription customers were slow to return.

Figure 3.1(a): Daily Order Volume Trend

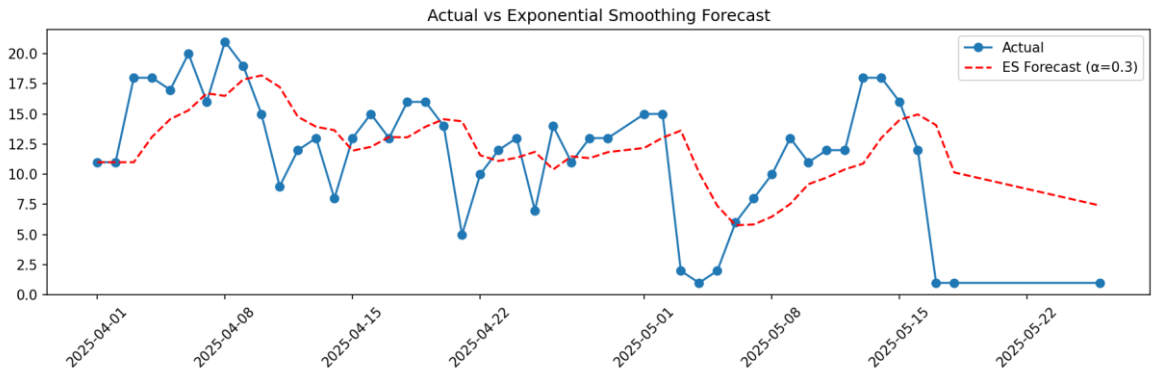

Figure 3.1(b):Actual vs Exponential Smoothing Forecast of Daily Order Volume Trend

3.2 Day-of-Week Demand Pattern

Analysis of 567 orders by day of week reveals a pronounced midweek peak that has direct implications for staffing and procurement scheduling. Thursday recorded the highest weekly order volume at 100 orders (17.64% of the total), followed closely by Tuesday with 92 orders (16.2%) and Wednesday with 83 orders (14.6%). The combined Wednesday–Thursday–Friday window accounted for 48.1% of all weekly orders. Monday registered the lowest volume at 57 orders (10.05%), representing a 75.4% gap between the weekly trough and peak. This pattern is highly consistent across the 7-week study period and should drive a structured procurement calendar: bulk non-perishable procurement on Mondays and bulk perishable procurement on Tuesdays to ensure full stock coverage for the Wednesday–Friday peak.

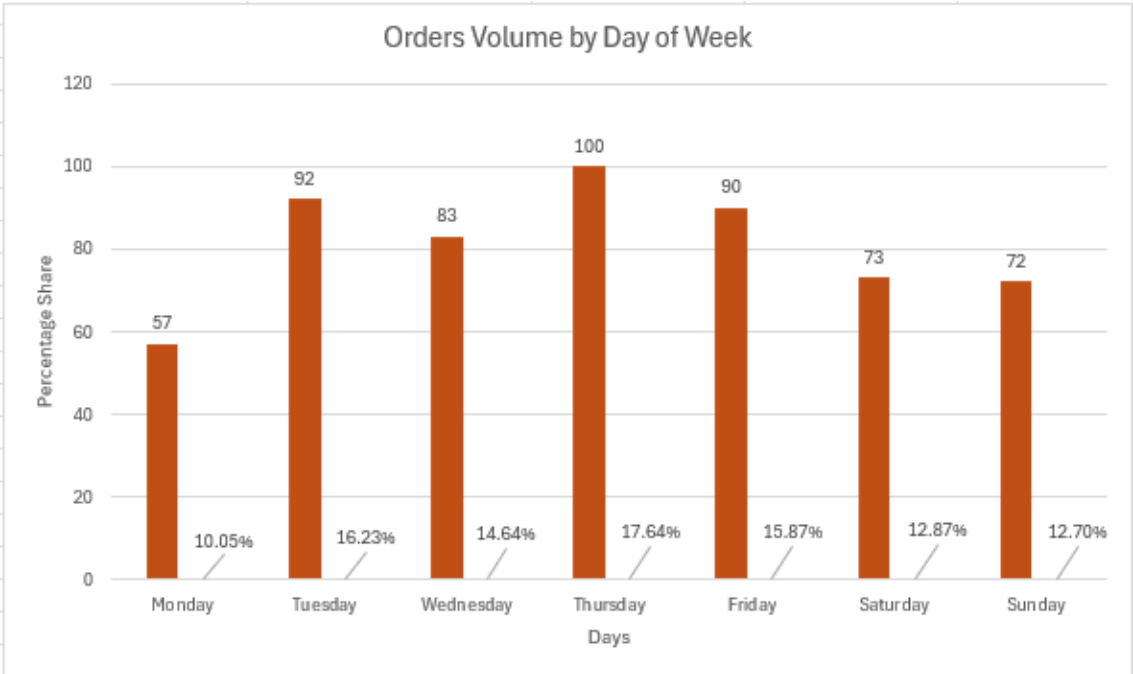

Figure 3.2: Order Volume by Day of Week

3.3 Order Type Distribution

Of the 567 total orders, 298 (52.6%) were subscription orders and 269 (47.4%) were regular orders. Despite subscription orders comprising the slight majority of volume, regular orders generated 68.8% of total revenue (Rs. 81,216) compared to only 31.2% (Rs. 36,770) for subscription orders. This is reflected in the Average Order Value: Rs. 302 for regular versus Rs. 123 for subscription. The day-of-week breakdown reveals an important pattern: Monday has the highest subscription concentration (66.7%), while Sunday has the highest regular order concentration (70.8%). This asymmetry suggests that subscription orders drive weekday predictability while regular orders drive weekend revenue. The 52.6%–47.4% split is a structurally balanced revenue model for a home kitchen business of this scale.

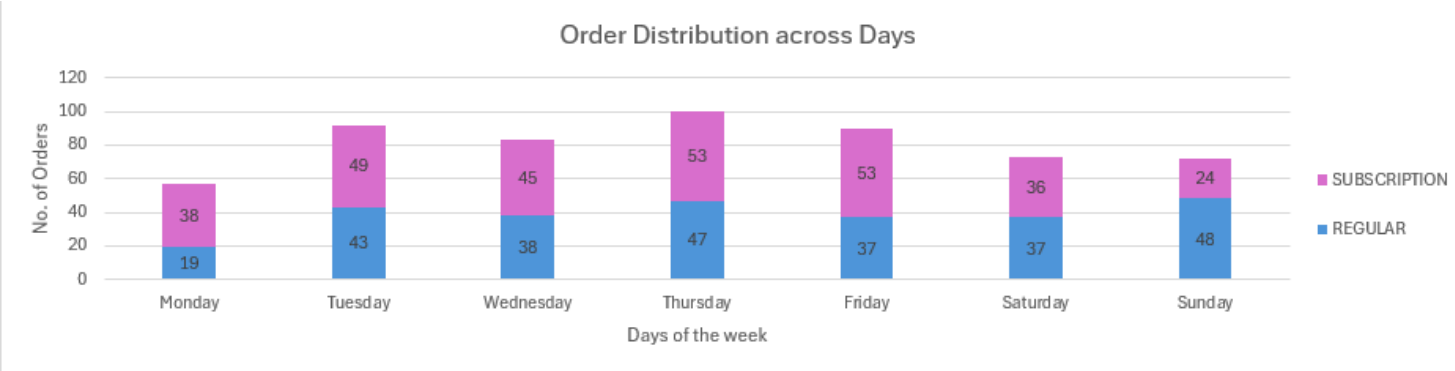

Figure 3.3: Subscription vs Regular Order Distribution

3.4 Meal Period Analysis

Lunch dominates order activity with 420 orders representing 74.1% of total volume, while generating an estimated 85.6% of total revenue at Rs. 100,951. Breakfast contributed 68 orders (12.0%) and dinner 79 orders (13.9%). This extreme concentration — nearly three-quarters of all orders in a single 2-hour lunch window — directly validates PS2 (peak-hour capacity strain). On Thursdays specifically, lunch orders reached 78 in a single day across the study period, representing the single highest-demand event observed. The industry benchmark for a balanced meal period distribution is approximately 40% lunch, yet this kitchen operates at nearly double that concentration, placing severe and recurring stress on Mrs. Padmavathy as the sole cook. Note that exact timestamps within the lunch window were not available in the dataset; this analysis is conducted at meal period granularity rather than hourly granularity.

Count of ORDER ID	Column Labels			
Row Labels	Breakfast	Dinner	Lunch	Grand Total
Monday	6	8	43	57
Tuesday	11	15	66	92
Wednesday	10	13	60	83
Thursday	12	10	78	100
Friday	13	16	61	90
Saturday	9	9	55	73
Sunday	7	8	57	72
Grand Total	**68**	**79**	**420**	**567**

Figure 3.4(a):Pivot Table: Meal Period Distribution

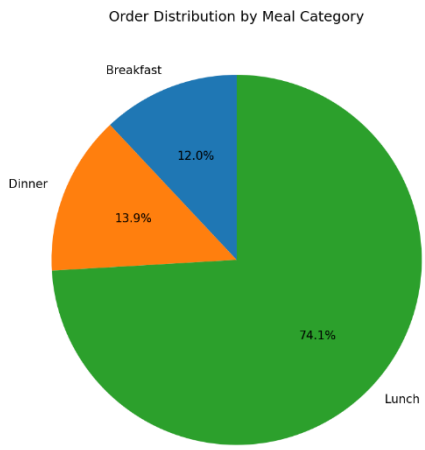

Figure 3.4(b): Meal Period Distribution Chart

3.5 Dish Popularity Analysis

Among regular orders, the top 10 dishes account for 78.4% of all regular order volume, with Keerai Paruppu Kadayal as the single most popular dish at 32 orders, followed by Chicken Biryani. This high concentration in a small set of dishes is strategically favourable: it means that a focused inventory strategy targeting just 10 dishes can effectively cover the vast majority of regular order fulfilment requirements. For subscription orders, the top 10 dishes account for only 28.9% of subscription volume, reflecting the deliberate variety built into subscription meal plans. This contrast has important procurement implications:

regular dish ingredients can be stocked in predictable bulk quantities, while subscription ingredient procurement must maintain broader coverage with smaller per-item quantities.

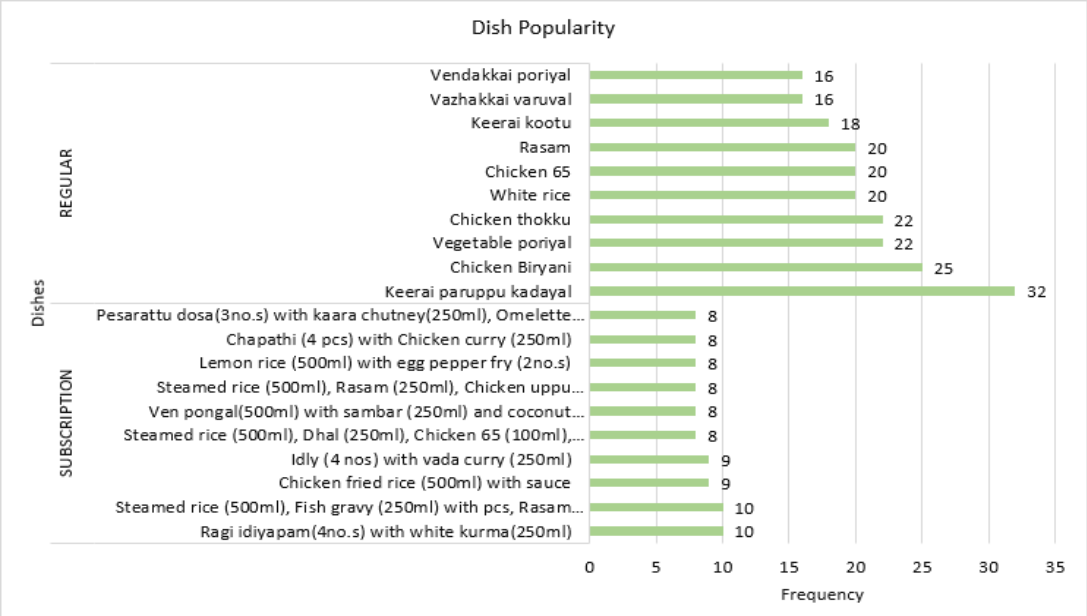

Figure 3.5: Dish Popularity by Order Type

3.6 Vegetarian vs Non-Vegetarian Analysis

Non-vegetarian dishes accounted for 317 orders (55.9% of total) while vegetarian dishes accounted for 250 orders (44.1%). Despite similar order volumes, non-vegetarian dishes contributed approximately 54% of total revenue, reflecting their higher average selling price. Chicken Biryani, the highest-revenue single dish, has a contribution margin of only 29.2% due to the high cost of Seeraga Samba rice and chicken at current local vendor prices. By contrast, vegetarian dishes such as Keerai Paruppu Kadayal (41.9% CM), White Rice (49.4% CM), and Vendakkai Poriyal (48.0% CM) offer substantially better margins, suggesting that promoting high-margin vegetarian combinations could improve overall profitability without reducing revenue.

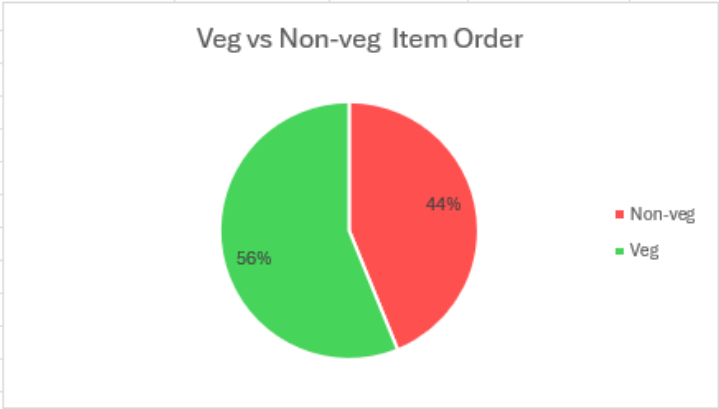

Figure 3.6: Veg vs Non-Veg Order Share

3.7 Revenue vs Order Count Trend

Daily revenue exhibited a coefficient of variation of 53.03%, which is notably higher than the order count CV of 44.44%. This divergence indicates that revenue volatility is driven not only by demand fluctuations but also by the mix shift between regular and subscription orders on any given day. On days with higher regular order concentration, revenue can be 2–3 times higher than on subscription-heavy days despite similar order counts. The 7-day kitchen closure in May 2025 resulted in a direct revenue loss of approximately Rs. 9,718.

Furthermore, the revenue recovery trajectory after reopening shows a 5–7 day lag, meaning the total economic cost of the closure, including foregone subscription revenue from customers who did not renew, likely exceeded Rs. 15,000.

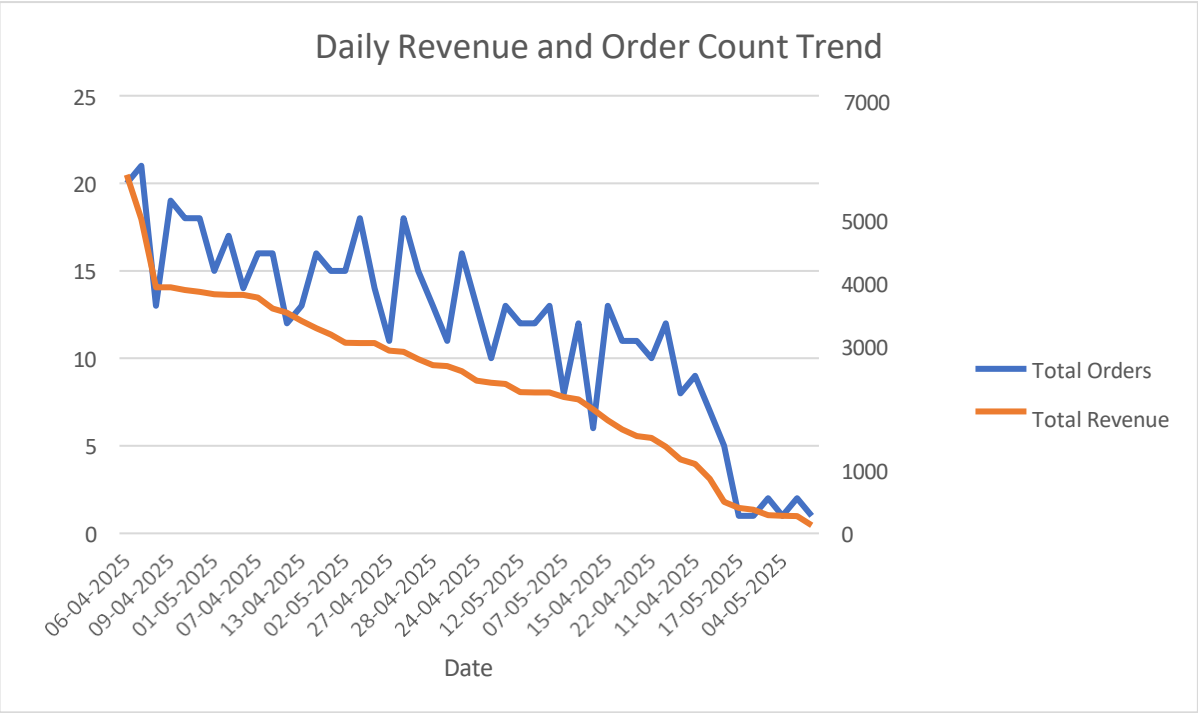

Figure 3.7: Daily Revenue and Order Count Trend

3.8 Statistical Distribution and Outlier Analysis

Box plot analysis of daily order volumes identifies the interquartile range (IQR) as 7 orders (Q1 = 8, Q3 = 15). The upper fence is 25.5 orders and the lower fence is –2.5 orders (effectively 0). Two categories of outliers are observed. Positive outliers (orders exceeding 21 per day) occurred on April 8, 2025, a festival date coinciding with elevated demand, and on two Thursdays in late April. These positive outliers represent peak capacity risk. Negative outliers (1–2 orders per day) are entirely explained by the kitchen closure period in May 2025. Table 3.1 provides the complete descriptive statistics.

Metric	Daily Orders	Daily Revenue (₹)
Mean	11.81	2,458
Median	12	2,390
Standard Deviation	5.25	1,303
Minimum	1	129
Maximum	21	5,736

Metric	Daily Orders	Daily Revenue (₹)
Q1 (25th percentile)	8	1,529
Q3 (75th percentile)	15	3,044
IQR	7	1,515
Upper Fence (Q3 + 1.5×IQR)	25.5	5,316
Lower Fence (Q1 − 1.5×IQR)	−2.5 (effective: 0)	−743 (effective: 0)
Coefficient of Variation	44.44%	53.03%

Table 3.1: Descriptive Statistics and Outlier Boundaries — Daily Orders and Revenue

4 Interpretation of Results and Recommendations

4.1 PS1 – Emergency Procurement Reduction

The demand forecasting analysis confirms that daily order volume is too volatile (CV = 44.44%) for intuition-based procurement. The Exponential Smoothing model achieves a MAPE of 25.51% when evaluated on operational days (excluding 6 closure days where orders fell to 1–2 due to illness rather than demand fluctuation). Combined with a ROP trigger system for 7 Class A ingredients, a structured weekly procurement routine can virtually eliminate emergency Quick Commerce purchases.

The recommended routine is as follows: every Sunday evening, Mrs. Padmavathy should review the 7-day ES forecast and compare each Class A ingredient's stock against its ROP, placing procurement orders for anything at or below the trigger. Monday should cover bulk non-perishables (White Rice, Toor Dal, Basmati Rice, Cooking Oil, Ghee — lead time 2–3 days), and Tuesday should cover perishables (Chicken, Coconut, Tomato, Drumstick — lead time 1 day) to ensure freshness for the Wednesday–Friday peak. A printed ROP reference card for the 7 Class A ingredients should be kept in the kitchen for daily reference.

The procurement cost-benefit analysis projects annual savings of Rs. 18,000–30,000 by eliminating Quick Commerce dependency for the five highest-markup ingredients (Tomato: 140%, Eggs: 133%, Chicken: 80%, Drumstick: 80%, Potato: 66.7%), reducing purchases from an estimated 3–4 per week to zero. This recommendation is Specific (ROP card for 7 ingredients), Measurable (weekly QC purchase count), Achievable (Mrs. Padmavathy has

already trialled this successfully for Toor Dal and White Rice), Relevant (directly addresses PS1), and Time-bound (full implementation within 4 weeks).

4.2 PS2 – Peak Hour Capacity Management

The data establishes that lunch constitutes 74.1% of all orders and Thursday alone accounts for 17.64% of weekly volume, making Thursday lunchtime a recurring capacity crisis. While the ROP trigger system addresses ingredient shortages, the staffing and order volume dimension requires direct intervention.

On the demand side, daily order caps should be introduced through the Cookr app — 15 orders on Thursday and Friday, 18 on Monday through Wednesday, and 20–22 on weekends. Once the cap is reached, the kitchen marks itself unavailable, preventing over-commitment without sacrificing revenue.

On the supply side, Mrs. Mariammal should be cross-trained to independently handle rice and sambar preparation, which constitute approximately 60% of all lunch components. Sambar and rice should be batch-prepared by 10:00 AM on Thursdays ahead of the 12:00–14:00 peak, and subscription components prepared on Sunday evenings for Monday delivery.

Target: Zero unplanned closures per calendar month within 3 months, measured by forced closures per month and Cookr platform rating.

4.3 PS3 – Menu Profitability Optimization

The Menu Engineering analysis of 20 dishes (10 regular, 10 subscription) identifies clear strategic actions for each quadrant. Among regular dishes, four Stars — Keerai Paruppu Kadayal (41.9% CM), Chicken Thokku (45.8%), White Rice (49.4%), and Chicken 65 (36.9%) — should receive zero-stockout priority in procurement planning and prominent placement in the Cookr app listing. Chicken Biryani, the second most popular dish, is classified as a Plowhorse at only 29.2% CM due to the high cost of Seeraga Samba rice and chicken. A price increase of Rs. 20 (from Rs. 190 to Rs. 210) would raise its CM to approximately 38%, converting it to a Star without likely impacting demand given its popularity.

Among subscription dishes, two Stars — Rice with Rasam and Chicken Uppu Kari (32.4% CM) and Chapathi with Chicken Curry (35.0% CM) — should be anchor dishes in the weekly subscription rotation. Ven Pongal with Sambar and Coconut Chutney is the most critical finding: at a selling price of Rs. 79, this dish generates a contribution margin of –Rs. 1.41, meaning Mrs. Padmavathy loses money on every portion sold after accounting for ingredient cost and platform fee. The immediate recommendation is to either raise the price to Rs. 99

(bringing CM to approximately 18.6%) or remove it from the subscription menu entirely and replace it with Idly with Vada Curry (37.0% CM), which is both more profitable and similarly popular.

Keerai Kootu (Dog, 21.4% CM) among regular dishes should be removed from the standalone menu and offered only as a combo accompaniment. Three Puzzle dishes — Rasam (37.8%), Vazhakkai Varuval (32.8%), and Vendakkai Poriyal (48.0%) — have good margins but low order frequency. These should be featured in app descriptions and offered as add-on upsell suggestions alongside high-popularity dishes. Target: Improve average CM ratio across all active dishes from the current 32% to 40% within 6 months. At current revenue levels (Rs. 1,17,986 for 49 days, approximately Rs. 28,900/month), an 8% improvement in average CM generates an additional Rs. 2,312 per month or Rs. 27,744 per year.

4.4 Phased Implementation Roadmap

Phase	Timeline	Actions	Target Metric
Immediate	Week 1–2	Print ROP reference card for 7 Class A ingredients; begin Mon/Tue bulk procurement routine; remove Ven Pongal from subscription menu	QC purchases per week = 0
Short-term	Month 1–2	Set up Sunday ES forecast review; introduce daily order caps in Cookr; cross-train helper for rice and sambar; replace Keerai Kootu with Vendakkai Poriyal in standalone menu	Zero forced closures/month
Medium-term	Month 3–6	Reprice Chicken Biryani to Rs. 210; feature Puzzle dishes as add-ons; consider Thu/Fri 10% busy-hour surcharge for regular orders	Avg CM ratio ≥ 38%

Phase	Timeline	Actions	Target Metric
Long-term	Month 6–12	Grow subscription base from 52.6% to 58% of orders; upskill helper for independent sambar preparation; evaluate second helper on Thursdays	Monthly revenue > Rs. 35,000

Table 4.1: Phased Implementation Roadmap

Total projected annual improvement: Rs. 18,000–30,000 (procurement savings, PS1) + Rs. 40,000 (avoided closure revenue loss, PS2) + Rs. 27,744 (CM improvement, PS3) = Rs. 85,744–97,744 per year. These projections are conservative and based solely on the 49-day dataset. Longer data collection across two full seasonal cycles would allow ARIMA modelling and more precise festival demand forecasting, further improving procurement accuracy.

5 Dataset Links and References

All datasets, analysis files, and proof of originality supporting this report are available at the following Google Drive location:

Google Drive Folder (Datasets, Excel Workbook, Colab Notebook, Output, Proof of Originality):

https://drive.google.com/drive/folders/1OnrAARNY5rJlQFLVmKC9WcsE1JYydXRr?usp=drive_link

The Google Drive folder contains the following files:

1. MASTER_MENU_SPREADSHEET.xlsx — Contains Order Data, Menu Details, Ingredient Costs, Operational Events, Availability Matrix, Descriptive Analysis, Demand Forecasting, ABC Classification, EOQ and ROP, and Contribution Margin sheets.

2. BDM_Colab_Analysis.ipynb — Google Colab notebook containing Python code for demand trend analysis, Exponential Smoothing forecast with MAPE computation, and all charts used in Section 3.

3. Proof_of_Originality — Folder containing kitchen operation photographs, Cookr app screenshots of orders and menu, 8-minute video interview with Mrs. Padmavathy Ravichandran, and printed authorization letter confirming consent for data use.